

Prototype Verification & Test Plan

Compact Scissor Lift Mechanism - Rev A

This plan defines the physical validation that would be required after manufacturing a prototype. No physical tests have been performed for the current portfolio release.

Test	Method	Acceptance criterion
Visual / dimensional inspection	Check main dimensions, hole positions, deburring, fit	Matches released drawing intent; no sharp burrs or interference
Unloaded motion test	Cycle from minimum to maximum height manually	Smooth motion; no binding; platform remains guided
Self-locking / back-drive check	Stop at several heights with no hand input	No unintended downward motion
Centered static load test	Apply 10 kg centered payload	No permanent deformation; stable support
Proof-load test	Apply controlled load above rated payload per a defined test procedure	No permanent deformation or instability; exact proof factor to be defined
Off-center stability check	Place controlled load at defined offsets	No tipping; travel remains controllable
Fastener retention check	Inspect after repeated cycles	No loosening or migration
Endurance cycling	Repeat full lift cycles under representative load	No progressive wear, binding, or damage
Post-test inspection	Inspect pins, holes, arms, screw and supports	No cracks, galling, excessive wear or permanent set

Safety note: This design is a bench-top portfolio concept. It is not intended or certified for lifting people. Any physical prototype should be tested in a controlled environment with appropriate guarding and supervision.